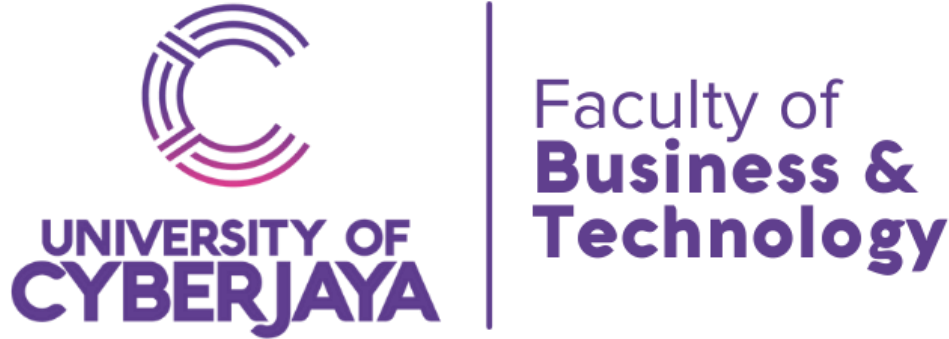


Cover page



FACULTY OF BUSINESS & TECHNOLOGY
ARTIFICIAL INTELLIGENCE (BIT 4543)
JULY 2026
ASSIGNMENT COVER SHEET

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Assignment Details

To be marked by: SHAHBIN AZUAN BIN NAZEER	
Date submitted:	Submission Due Date: 4 th September
Submission of this assignment agrees to the following: <i>I understand that the piece of work submitted will be considered as the final and complete version of my assignment of which I am the sole author. I understand both the meaning and consequences of plagiarism and that my work has been appropriately attributed unless otherwise stated. I have not knowingly allowed another to copy my work.</i>	

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Declaration and Acknowledgement

This report is the original work of the group except where sources are acknowledged. The group accepts responsibility for the accuracy of the work, results, references and screenshots included in the final submission
Group Leader's Signature: <u><i>Danish</i></u> Group Leader's Name: Danish Mohd Yunus Group Leader's Matrix ID: 2503-0850

Abstract

The rapid growth of e-commerce has resulted in a large volume of customer-generated reviews containing valuable information about product experiences, satisfaction, and opinions. However, manually analysing thousands of reviews is time-consuming and difficult to scale. This project develops an artificial intelligence-based sentiment analysis system for Amazon product reviews that automatically classifies reviews as positive, neutral, or negative. The system uses the Amazon Reviews'23 dataset and focuses on two product categories: Video Games and All Beauty. A target sample of 50,000 reviews was prepared for each category. Sentiment labels were derived from star ratings, with one- and two-star reviews labelled negative, three-star reviews labelled neutral, and four- and five-star reviews labelled positive. After cleaning, the final datasets contained 48,371 Video Games reviews and 48,443 All Beauty reviews. Natural Language Processing techniques were applied, including HTML and URL removal, stop-word removal, preservation of important negation words, and WordNet lemmatisation. The cleaned reviews were converted into numerical features using Term Frequency-Inverse Document Frequency (TF-IDF). A Majority Class baseline, Multinomial Naive Bayes, and Logistic Regression were evaluated using an 80/20 stratified train-test split. Logistic Regression was selected for deployment and achieved 68.34% accuracy and a 68.37% weighted F1-score for Video Games, and 69.33% accuracy and a 69.28% weighted F1-score for All Beauty. The models were integrated into an interactive Streamlit dashboard for individual predictions, dataset exploration, model-performance analysis, history, and feedback. User acceptance testing recorded 25 of 27 tasks completed successfully, representing a 92.6% completion rate. The project demonstrates that a transparent and computationally efficient machine-learning approach can provide a practical prototype for analysing large collections of customer reviews, while still requiring human judgement for mixed, sarcastic, or informal language.

1.0 Introduction

1.1 Background

Online shopping platforms generate substantial amounts of customer feedback through product reviews. These reviews provide information about customers' experiences, satisfaction and opinions that can help businesses understand how customers perceive their products. However, the increasing number of reviews makes it difficult for organisations to manually examine every review.

Sentiment analysis provides a method for automatically identifying the overall opinion expressed in textual data. It is an application of Natural Language Processing (NLP) in which text is processed and classified according to sentiment. Sentiment analysis is commonly applied to customer reviews because automated classification can process substantially larger volumes of information than manual review. Research has investigated a wide range of sentiment-analysis techniques, including lexicon-based approaches, traditional machine-learning algorithms, deep learning and transformer-based models. A systematic review by Adak et al. identifies machine-learning and deep-learning approaches as major approaches to sentiment classification and highlights the importance of explainability when applying complex AI models.

For e-commerce platforms such as Amazon, sentiment analysis is particularly useful because customer reviews contain direct feedback about products. Recent research has specifically investigated sentiment analysis on Amazon reviews using techniques ranging from TF-IDF and traditional machine learning to Fast Text, deep learning and transformer-based models. Hashmi and Mayilan demonstrate the wide range of approaches that can be applied to Amazon review sentiment analysis, while also highlighting the value of incorporating contextual product information.

1.2 Problem Statement

Amazon contains many customer reviews across many product categories. Although these reviews provide useful information, the volume of textual data makes manual analysis inefficient. Further challenge is that customer reviews are unstructured. Customers may use informal language, abbreviations, spelling variations, punctuation, HTML elements and different expressions of opinion. Therefore, textual reviews must be cleaned and converted into a numerical representation before they can be processed by a machine-learning model.

Another challenge is the selection of an appropriate classification algorithm. More complex deep-learning and transformer models may provide strong predictive performance but can require greater computational resources and introduce additional implementation complexity. Research has shown that traditional linear models can still be effective for large-scale customer-review classification. Chinnalagu and Durairaj, for instance, investigated machine-learning linear models for customer-review sentiment analysis and demonstrated the continued relevance of efficient traditional approaches.

Therefore, this project investigates whether a practical TF-IDF and Logistic Regression approach can provide useful sentiment classification for Amazon product reviews while remaining sufficiently simple to deploy as an interactive application.

1.3 Project Aim

The main aim of this project is to develop and evaluate a machine-learning-based sentiment analysis system capable of automatically classifying Amazon product reviews into positive, neutral and negative sentiment categories.

1.4 Project Objectives

1. To develop an AI-based application that classifies Amazon review text as negative, neutral or positive.
2. To collect and clean a large working sample of Amazon reviews from the Video Games and All Beauty categories.
3. To train category specific TF-IDF and Logistic Regression sentiment models.
4. To visualise sentiment distributions, rating, review lengths and model performance results.
5. To provide a practical interface for entering reviews and recording optional user feedback.
6. To evaluate the model with classification metrics and evaluate the application through user acceptance testing.

1.5 Scope of Project

The system analyses English-language review data from the Video Games and All Beauty categories. The working datasets have a target samples sizes of 50,000 records per category; after cleaning and deduplication, 48,371 Video Games reviews and 48,443 All Beauty reviews remain. Sentiment labels are derived from star rating; 1-2 stars are negative, 3 stars are neutral and 4-5 stars are positive.

Moreover, the application includes category selection, review text prediction, probability display, dataset charts, model metrics, analysis history and feedback recording. However, the dashboard do not include user login, real-time Amazon data collection, aspect-based sentiment explanation, multilingual model or automated business decision. File upload is outside the current prototype scope unless it is added and tested separately.

1.6 Significance of the Project

The project demonstrates how an organization could implement NLP to obtain high-level view of customer opinion. A sentiment dashboard can help users identify categories with more negative feedback, inspect changes in rating distributions and select reviews for manual investigation. The system also supports leaning outcomes by connecting data preparation, supervised learning, evaluation, interface design and database integration in one application.

2.0 Literature Review

2.1 Overview to Artificial Intelligences Technologies

Artificial Intelligence (AI) refers to the development of computer systems that can perform tasks normally associated with human intelligence, such as learning, reasoning, classification, and decision-making. In this project, AI is applied to analyse customer review text and identify the sentiment expressed by the reviewer. Machine Learning (ML) is a branch of AI in which algorithms learn patterns from existing data. Instead of manually defining every possible rule, a machine-learning model is trained using examples containing review text and sentiment labels. This project uses supervised machine learning because the review records contain predefined sentiment labels derived from their star ratings.

Deep Learning is a subfield of machine learning that uses multi-layer neural networks to learn complex patterns. Deep-learning models such as recurrent neural networks, convolutional neural networks, and transformers can capture more complicated language relationships. However, these models generally require more computational resources, larger training datasets, and more complex tuning. Therefore, deep learning was considered as a possible future improvement but was not selected for the current prototype.

Natural Language Processing (NLP) allows computers to process and analyse human language. Sentiment analysis is an NLP classification task that determines whether a text expresses a positive, neutral, or negative opinion. In this project, NLP preprocessing is used to clean the review text before it is converted into numerical features using Term Frequency-Inverse Document Frequency (TF-IDF).

TF-IDF assigns higher importance to words that are frequent in a particular review but less common across the full collection of reviews. The resulting numerical representation is passed to a Logistic Regression classifier. This approach was selected because it is computationally efficient, relatively easy to interpret, and suitable for high-dimensional text data.

2.2 Related Works

Early sentiment-analysis research established that opinions can be represented as a text-classification problem. Pang et al. (2002) demonstrated that machine-learning algorithms could identify sentiment from written reviews. Their work provided an important foundation for supervised sentiment classification, although the datasets and feature representations used at that time were smaller and less diverse than modern e-commerce datasets.

Wang and Manning (2012) showed that simple unigram and bigram features can produce strong results for sentiment and topic classification. Their findings are relevant to this project because the proposed system also uses word-level and two-word phrase-level features. This supports the use of TF-IDF as a transparent baseline before more complex contextual models are considered.

Chinnalagu and Durairaj (2021) studied customer-review sentiment using FastText and a Linear Support Vector Machine, with comparison against a bidirectional LSTM approach. Their research demonstrates that efficient feature-based and linear models can be useful for large-scale customer-review classification. However, their use of FastText and Linear SVM differs from the current project, which uses TF-IDF and Logistic Regression. This difference illustrates why model performance should only be compared when the datasets, labels, and evaluation procedures are equivalent.

Hashmi and Yayilgan (2025) proposed a product-context-aware and explainable approach for Amazon reviews. Their work incorporated additional product information and compared several machine-learning, deep-learning, and transformer-based techniques. The study also applied explainability techniques to help interpret model decisions. This is relevant to the current project because the meaning of a sentiment can depend on the product domain. However, the approach is more complex than the current prototype, which focuses on overall review sentiment rather than product aspects.

Vadla et al. (2024) developed a BERT- and T5-based pipeline for identifying product aspects and sentiments in Amazon reviews. Their approach used contextual transformer models to provide product-design insights.

This demonstrates the advantages of advanced NLP models when reviews contain multiple aspects or mixed opinions. Nevertheless, the approach requires more computational resources and more specialised training data than the current TF-IDF and Logistic Regression pipeline.

The main findings from the related studies are summarised below.

Study	Main approach	Contribution	Limitation in relation to this project
Pang et al. (2002)	Supervised sentiment classification	Established sentiment analysis as a machine-learning task	Older datasets and feature representations
Wang and Manning (2012)	Unigram and bigram text features	Demonstrated that simple text features can be effective	Limited contextual understanding
Chinnalagu and Durairaj (2021)	FastText, Linear SVM, and BiLSTM	Compared efficient linear and deep-learning methods for reviews	Uses a different feature representation and evaluation setting
Hashmi and Yayilgan (2025)	Product-context-aware learning and explainable AI	Includes product context and model interpretability	More complex and computationally demanding
Vadla et al. (2024)	BERT and T5 for aspect and sentiment analysis	Captures richer contextual and aspect-level information	Requires advanced models and additional training resources

The literature indicates that there is a trade-off between model complexity, contextual understanding, interpretability, and computational cost. Traditional linear models are easier to deploy and explain, while transformer models may provide stronger contextual understanding but require greater resources.

2.3 Research Gap

First, studies use different sentiment definitions. Some use binary sentiment labels, while others use multi-class or aspect-level labels. Some datasets use manually annotated opinions, whereas the current project uses star ratings as a proxy for sentiment. This makes direct comparison of reported accuracy values inappropriate.

Second, many studies focus primarily on model performance and do not provide an end-to-end application for non-technical users. A prediction model may achieve good results in an experiment but still be difficult to use without an interface, visual summaries, and clear explanations.

Third, advanced contextual models may be unsuitable for a small academic deployment because of their computational and implementation requirements. There is therefore a practical need for a transparent baseline that can be trained, evaluated, deployed, and reproduced using accessible tools.

This project addresses these gaps by developing a complete sentiment-analysis workflow consisting of category-specific datasets, rating-based three-class labels, TF-IDF features, Logistic Regression models, model comparison, a Streamlit interface, dataset visualisation, analysis history, and user feedback. The contribution is a practical and reproducible AI application rather than a claim of developing a new classification algorithm.

The project also makes the limitations of the label design explicit. A star rating represents the reviewer's overall evaluation, but the text may contain both positive and negative comments. For example, a review may praise a product's design while criticising its durability. Consequently, some reviews may be classified as neutral or may be difficult for the model to interpret correctly.

3.0 Methodology

3.1 Development Methodology

This project followed an iterative data-science development methodology, whereby development process was divided into several stages: requirement identification, dataset preparation, model development, application development, system testing, and evaluation.

First, the project requirements were identified based on the need for an interactive system that could classify customer reviews into positive, neutral, and negative sentiment categories. The datasets were then collected and prepared for analysis. This included selecting reviews from the Video Games and All Beauty categories, assigning sentiment labels from star ratings, removing unsuitable records, and cleaning the review text. After the data preparation stage, different machine-learning models were trained and compared. TF-IDF was used to convert review text into numerical features, while Logistic Regression and Multinomial Naive Bayes were evaluated against a Majority Class baseline. The best-performing model was selected for integration into the dashboard.

Lastly, application was developed using Streamlit. The dashboard was tested through functional testing and user testing. Feedback from testers was used to identify usability issues and confirm whether the main features operated as expected. This iterative process allowed the system to be improved progressively throughout development.

3.2 System Architecture

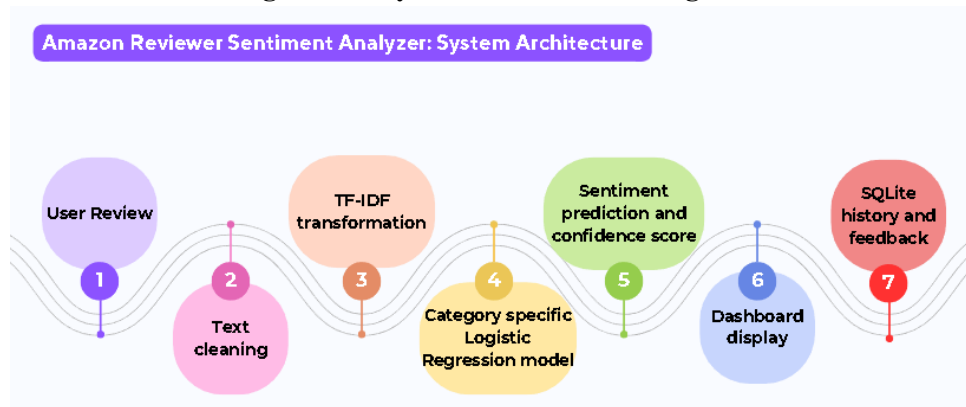
The system consists of five main layers: the data layer, preprocessing layer, machine-learning layer, application layer, and storage layer. These data layer contains the processed Amazon review datasets for the Video Games and All Beauty categories. Where each category has its own dataset and trained sentiment model. The preprocessing layer cleans the text using the same procedure used during model training. This ensures that new reviews entered by users are processed consistently with the training data.

The machine-learning layer contains a TF-IDF vectorizer and a Logistic Regression model for each category. When a user selects a category, the corresponding vectorizer and model are loaded. The submitted review is cleaned, transformed into TF-IDF features, and passed to the Logistic Regression model. Next, the model then predicts one of three sentiment classes: positive, neutral, or negative. The model probability is also displayed as an indication of prediction confidence.

The application layer is implemented using Streamlit and provides the Review Analyser, Dataset Dashboard, Model Performance, History and Feedback, and Methodology pages. The Review Analyser allows users to enter a review and receive a prediction. The Dataset Dashboard displays sentiment distributions and rating patterns. The Model Performance page displays classification metrics and confusion matrices.

The storage layer uses a local SQLite database in which submitted analyses are stored with the selected category, review text, predicted sentiment, confidence score, and timestamp. Users can also submit feedback regarding whether the prediction was correct. The database is used for the coursework demonstration and does not replace the trained machine-learning models.

Figure 3.1: System Architecture Diagram



3.3 Data Collection

The project used Amazon Rewviews'23 dataset in which customer-review data for two product categories: Video Games and All Beauty. These categories were selected to demonstrate how sentiment classification can be applied to different product domains. The raw review data contained several fields, including the star rating, review title, review text, product identifier, timestamp, helpful-vote count, and verified-purchase indicator. The raw files were processed into CSV format for use in the data-preparation and modelling stages.

A target sample of approximately 50,000 reviews was prepared for each category. The sampling process attempted to maintain a balanced distribution of the three sentiment classes. A fixed random seed was used to make the sampling process reproducible.

Star ratings were converted into sentiment labels using the following rule:

Rating	Sentiment label
1 or 2 stars	Negative
3 stars	Neutral
4 or 5 stars	Positive

After duplicate records, missing text, and empty cleaned reviews were removed, the final datasets contained the following numbers of records:

Category	Final records	Negative	Neutral	Positive
Video Games	48,371	16,495	16,316	15,560
All Beauty	48,443	16,382	16,260	15,801

The processed datasets retained the original review information and included an additional `clean_text` field. The `clean_text` field was used as the input for model training and prediction.

3.4 Data Preprocessing

Data preprocessing was performed to improve the quality and consistency of the review text before machine-learning training. The same cleaning procedure was applied to both the training data and reviews entered through the dashboard.

The preprocessing steps were as follows:

1. Rows with missing review text were removed.
2. Duplicate reviews were removed using the review text.
3. HTML tags and line-break elements were removed.
4. Text was converted to lowercase.
5. URLs and web addresses were removed.
6. Apostrophe and encoding artefacts were normalised.
7. Punctuation, numbers, and other non-alphabetic characters were removed.
8. Extra whitespace was reduced to a single space.
9. Stop words were removed to reduce unnecessary terms.
10. Important negation words, such as “not”, “no”, “never”, and “nothing”, were retained because they can change the meaning of a review.
11. Words containing two or fewer characters were removed.
12. Remaining words were lemmatised using the WordNet lemmatiser.
13. Reviews that became empty after cleaning were excluded.

For example, a review containing HTML tags, punctuation, and unnecessary stop words was converted into a shorter representation containing the main sentiment-related terms. The original text column was retained for reference, while the cleaned version was stored in the `clean_text` column. The sentiment label was assigned before model training using the star-rating mapping described in Section 3.3. This produced three classification categories: negative, neutral, and positive.

3.5 AI Model Development

The project used supervised machine learning for three-class text classification. The input variable was the cleaned review text, while the target variable was the sentiment label. For each category, the data was divided into training and testing sets using an 80:20 split. Stratified sampling was applied so that the proportions of negative, neutral, and positive reviews remained similar in both sets. The random state was fixed at 42 to ensure that the experiment could be reproduced.

TF-IDF Feature Extraction

Because machine-learning algorithms cannot directly process raw text, the review text was converted into numerical features using Term Frequency-Inverse Document Frequency, commonly known as TF-IDF.

The TF-IDF vectorizer was configured with the following settings:

- Maximum features: 30,000
- N-gram range: unigrams and bigrams
- Minimum document frequency: 2
- Maximum document frequency: 0.95
- Sublinear term-frequency scaling: enabled

Unigrams represent individual words, while bigrams represent pairs of consecutive words. Including both allowed the model to learn individual sentiment terms as well as short phrases such as “not good” or “highly recommended”.

The vectorizer was fitted only on the training data and the testing data was transformed using the already fitted vectorizer. This prevented information from the testing set from influencing the feature vocabulary and reduced the risk of data leakage.

Logistic Regression

Logistic Regression was selected as the main deployed model because it is suitable for high-dimensional text features and provides class probabilities for prediction confidence. A separate Logistic Regression model was trained for each product category. The model was configured with a maximum of 1,000 iterations and a fixed random state of 42. During prediction, the model returns one of the three sentiment labels and the probability associated with each class.

Model Comparison

Two additional approaches were used for comparison:

- Majority Class baseline
- Multinomial Naive Bayes
- Logistic Regression

The Majority Class baseline always predicts the most common sentiment class and provides a minimum reference performance. Multinomial Naive Bayes was included because it is a commonly used algorithm for text classification. Logistic Regression was selected as the final model after comparing the classification results.

The models were evaluated using accuracy, precision, recall, weighted F1-score, and confusion matrices. Separate models were maintained for Video Games and All Beauty rather than combining both categories into one model. This allowed each model to learn vocabulary and sentiment patterns specific to its product domain.

3.6 Tools and Technologies

Component	Technology used
Programming language	Python
User interface	Streamlit
Data processing	Pandas
Machine-learning framework	Scikit-learn
Text processing	NLTK
Feature extraction	TF-IDF Vectorizer
Main classification model	Logistic Regression
Comparison model	Multinomial Naive Bayes
Model storage	Joblib
Application database	SQLite
Data format	CSV and JSON Lines
Version control	Git and GitHub

Streamlit was used to develop the interactive dashboard because it supports development of Python-based data applications. Pandas was used to load and analyse the datasets, while scikit-learn was used for feature extraction, model training, prediction, and evaluation. NLTK was used for stop-word removal and lemmatisation. SQLite was selected for storing local analysis history and user feedback.

4.0 System Design and Implementation

4.1 System Requirement

4.1.1 Functional Requirements

Requirement	Functional requirement
FR1	The user can select either the Video Games or All Beauty category.
FR2	The user can enter an Amazon product review.
FR3	The system can classify the review as positive, neutral, or negative.
FR4	The system displays the prediction confidence for each sentiment class.
FR5	The user can view dataset statistics and sentiment distributions.
FR6	The user can view model accuracy, precision, recall, F1-score, and confusion matrices.
FR7	The system can save submitted analyses to the analysis history.
FR8	The user can submit feedback about whether a prediction was correct.
FR9	The user can view the methodology and limitations of the system.
FR10	The user can clear locally stored analysis history after confirmation.

4.1.2 Non-Functional Requirements

- The interface should be simple enough for a non-technical user.
- The application should respond to a review without unnecessary delay.
- The same text-cleaning process should be used during training and prediction.
- The application displays a warning when the review is blank or contains no meaningful words.
- Model and dataset loading should be cached to avoid repeated processing.
- The model evaluation should be reproducible using a fixed random seed.
- The system should validate database input and prevent invalid sentiment values.
- Users should be informed not to enter confidential or personally identifiable information.

4.2 Use Case

The main actor in the system is the **User**. The user interacts with the dashboard through the following use cases:

- Select product category
- Enter a review
- Analyse sentiment
- View prediction confidence
- View the Dataset Dashboard
- View Model Performance
- View Methodology
- View analysis history
- Submit prediction feedback
- Clear local history

The user first selects either the Video Games or All Beauty category. The category determines which dataset, TF-IDF vectorizer, and Logistic Regression model are used. The user can then enter a review and request a prediction. After the prediction is displayed, the user may review the saved analysis and provide feedback.

Figure 4.2.1: User Case Diagram



4.3 Activity Diagram

The main activity begins when the user opens the Streamlit application. The user selects a product category and chooses either to analyse an individual review or explore one of the dashboard pages. For an individual prediction, the user enters a review and selects the analysis button. The system first checks whether the input contains meaningful text. If the input is empty, the system displays a warning and requests another review. If the input is valid, the system cleans the text, loads the selected category's TF-IDF vectorizer and Logistic Regression model, and transforms the review into numerical features. The classifier then predicts the sentiment and calculates the probability for each class. Predicted sentiment and confidence values are displayed to the user. The analysis is also stored in the local SQLite database when database storage is available. The user can later open the History and Feedback page, select a saved analysis, and indicate whether the prediction was correct. If the prediction was incorrect, the user can provide a corrected sentiment and an optional comment.

4.4 Database Design

SQLite was selected as the database because the project requires a small local storage system for the coursework demonstration. The database stores analysis history and optional user feedback. Hence, the database contains two tables: analyses and feedback.

Analyses Table

The analyses table stores each submitted review and its prediction.

Field	Description
id	Unique analysis identifier and primary key
created_at	Date and time when the analysis was saved
category	Selected product category
review_text	Original review entered by the user
predicted_sentiment	Predicted positive, neutral, or negative class
confidence	Probability associated with the predicted class

Feedback Table

The feedback table stores the user's evaluation of a saved prediction.

Field	Description
id	Unique feedback identifier and primary key
analysis_id	Foreign key referring to the related analysis
prediction_correct	Indicates whether the prediction was correct
corrected_sentiment	Corrected class when the prediction was incorrect
comment	Optional user comment
created_at	Date and time when the feedback was submitted

Each analysis can have zero or one feedback record. The analysis_id field is unique in the feedback table, preventing multiple feedback records from being created for the same analysis. A foreign-key relationship with cascading deletion is used so that related feedback is removed when the associated analysis is deleted.

4.5 User Interface Design

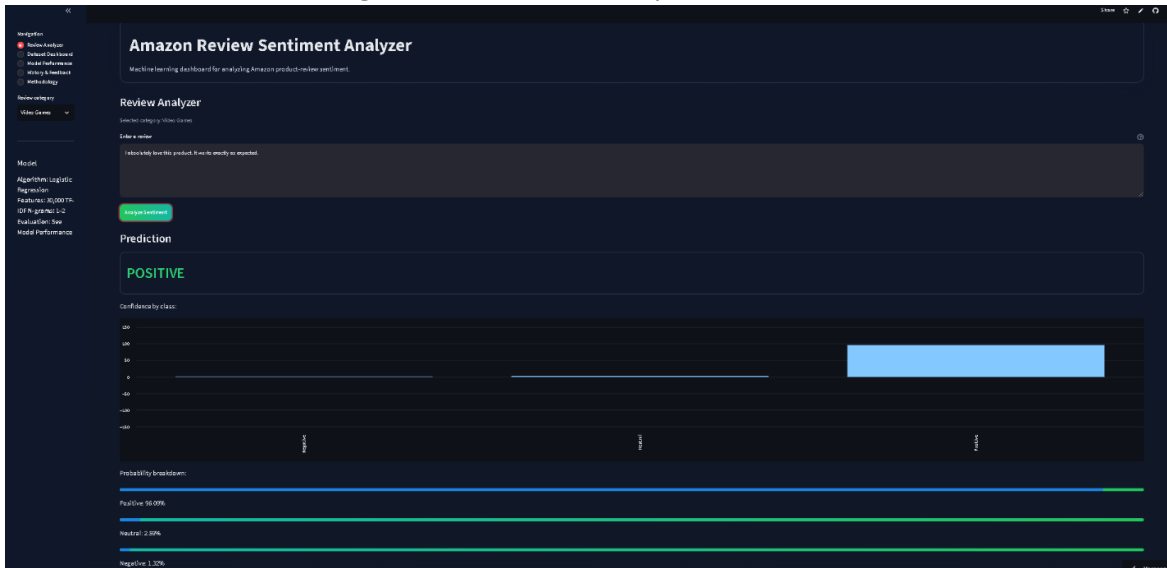
Review Analyzer

The Review Analyzer is the main page of the application. It contains:

- A product-category selector
- A text area for entering a review
- An analysis button
- The predicted sentiment
- A confidence chart
- The probability of each sentiment class
- A message indicating whether the analysis was saved successfully

The selected category determines which trained model is used. For example, selecting Video Games loads the Video Games TF-IDF vectorizer and Logistic Regression model.

Figure 4.4.1: Review Analyzer Screenshot



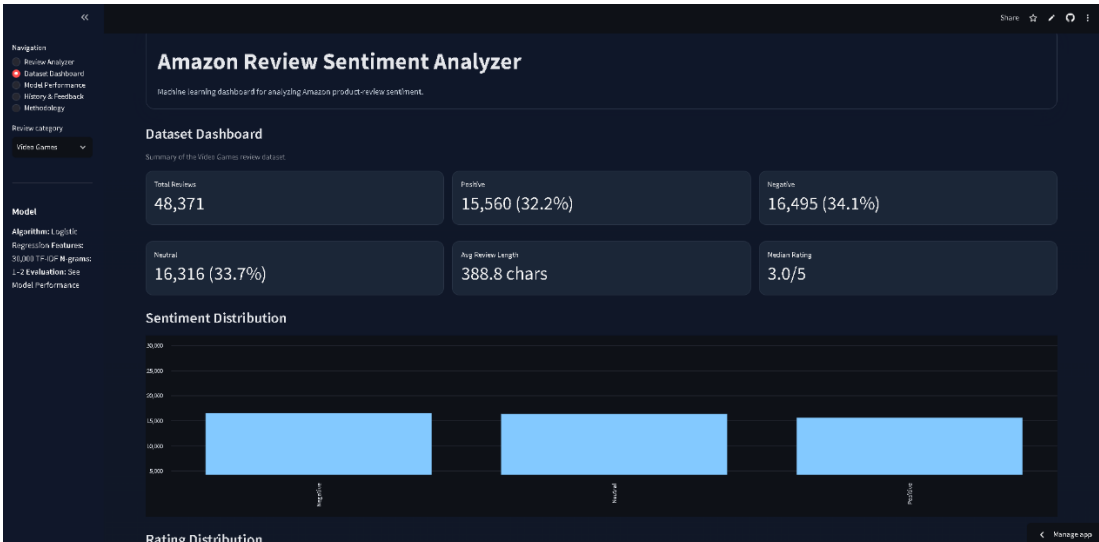
Dataset Dashboard

The Dataset Dashboard provides an overview of the selected category. It displays:

- Total number of reviews
- Positive, neutral, and negative review counts
- Sentiment percentages
- Star-rating distribution
- Verified-purchase distribution
- Sentiment by star rating
- Review-length information

These visualisations help users understand the overall sentiment patterns in the dataset.

Figure 4.4.2: Dataset Dashboard Screenshot



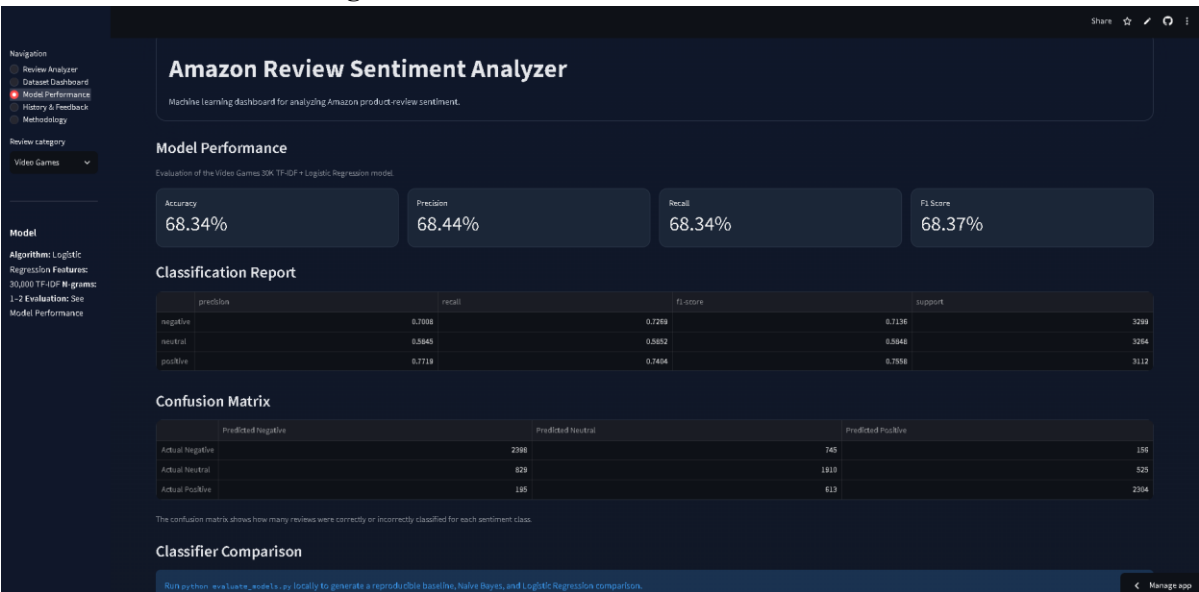
Model Performance

The Model Performance page displays:

- Accuracy
- Precision
- Recall
- F1-score
- Classification report
- Confusion matrix
- Model-comparison results

The page allows users to inspect how well the deployed model performs for the selected category. It also shows the comparison between the Majority Class baseline, Multinomial Naive Bayes, and Logistic Regression.

Figure 4.4.3: Model Performance Screenshot



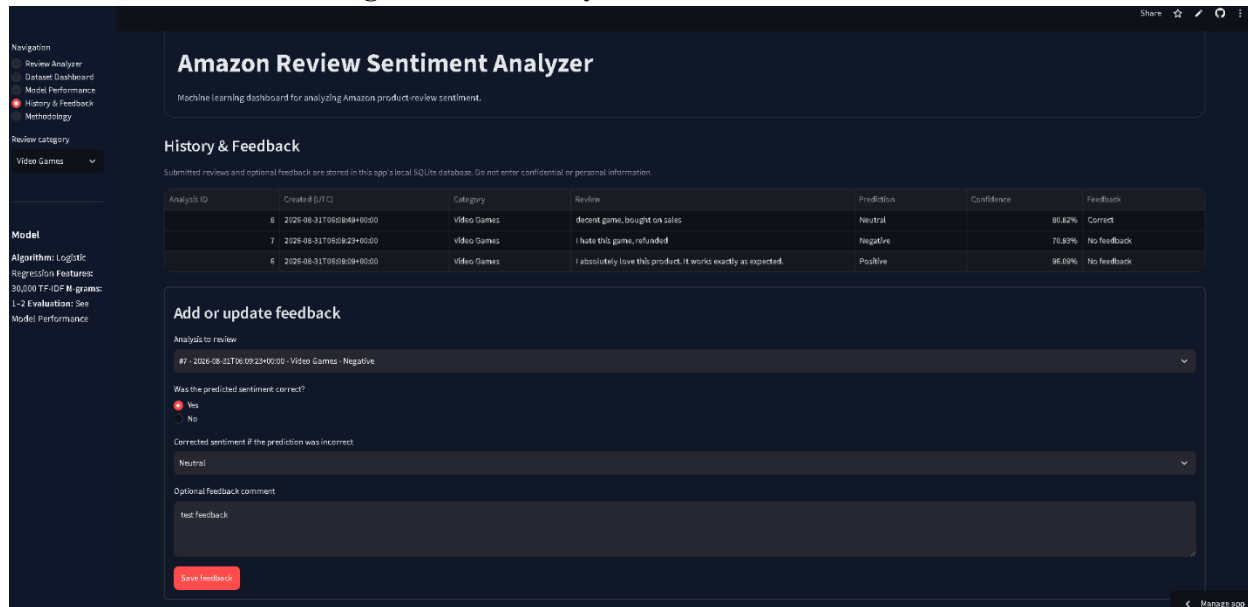
History and Feedback

The History and Feedback page displays recent analyses, including:

- Category
- Shortened review text
- Predicted sentiment
- Confidence
- Date and time
- Feedback status

Users can select a saved analysis and submit feedback. If a prediction is incorrect, the user can select the corrected sentiment and provide a comment.

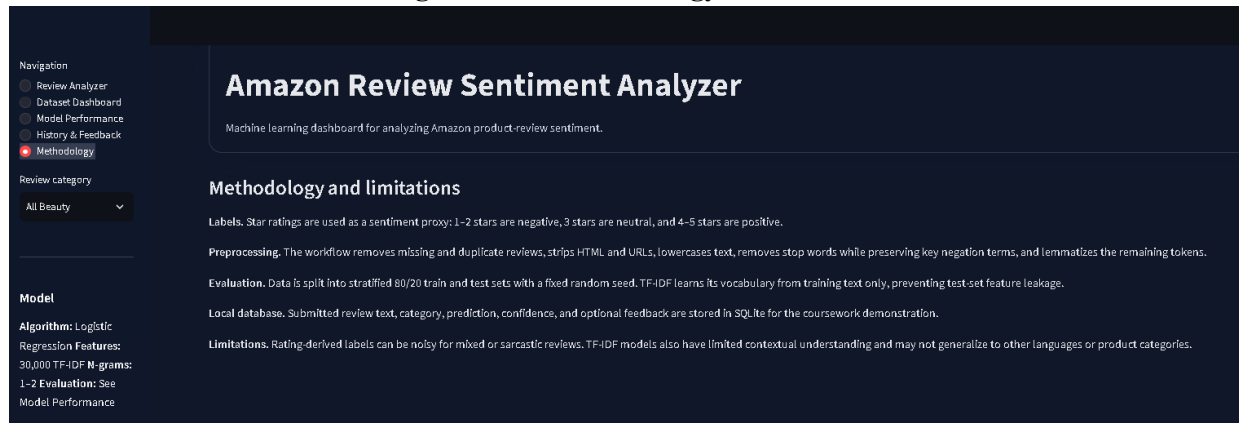
Figure 4.4.4: History and Feedback Screenshot



Methodology

The Methodology page explains:

- The star-rating-to-sentiment mapping
- Text preprocessing
- TF-IDF feature extraction
- Logistic Regression
- Evaluation procedure
- Database use
- System limitations

Figure 4.4.5: Methodology Screenshot

4.6 System Implementation

The main application is implemented in Python. The `app.py` file controls the Streamlit interface, page navigation, model loading, prediction process, dashboard charts, and feedback functions.

The category configuration connects each product category to its corresponding dataset, TF-IDF vectorizer, and Logistic Regression model. When a category is selected, the required model files are loaded using Joblib. Dataset and model loading are cached to reduce unnecessary processing during Streamlit reruns.

The review-prediction workflow performs the following steps:

1. Receive the review entered by the user.
2. Check whether the review contains meaningful text.
3. Remove HTML tags, URLs, punctuation, and unnecessary characters.
4. Convert the text to lowercase.
5. Remove stop words while preserving important negation words.
6. Lemmatise the remaining words.
7. Transform the cleaned review using the selected TF-IDF vectorizer.
8. Generate a prediction using Logistic Regression.
9. Calculate the probability of each sentiment class.
10. Display the result in the dashboard.
11. Save the analysis to SQLite when storage is available.

The `prepare_and_train_category.py` script performs balanced sampling, rating-based sentiment labelling, text cleaning, TF-IDF training, and Logistic Regression training. The `evaluate_models.py` script compares the baseline and machine-learning models using the same stratified train-test split. The `history_store.py` module handles database creation, analysis storage, feedback storage, history retrieval, and history deletion.

Input validation is used throughout the system. Blank reviews are rejected before prediction. The database layer validates category names, sentiment labels, confidence values, review length, and feedback length. Parameterised SQL statements are used when inserting or updating records.

4.7 Deployment

The application is deployed using Streamlit Community Cloud from the project's GitHub repository. The deployment includes the Streamlit application, processed datasets, trained model files, and required Python packages. The raw downloaded review files are excluded from version control because of their size. The local SQLite database is also excluded because it contains runtime history and feedback rather than permanent training data.

Streamlit application URL: <https://ai-sentiment-analysis-zz8emsybtgbi8rj3ltc57s.streamlit.app/#review-analyzer>

Final Git commit: <https://github.com/minerb388-wq/AI-Sentiment-Analysis>

5.0 System Implementation

5.1 Testing Strategy

Testing was conducted to verify both the technical behaviour of the system and its usefulness to end users. The testing process consisted of automated unit testing, functional testing, integration testing, model evaluation, and user acceptance testing.

Unit testing was used to verify individual functions such as sentiment-label mapping, text cleaning, balanced sampling, database recording, feedback updates, and input validation. Functional testing checked whether users could navigate the application and use the main dashboard features. Integration testing checked the connection between the user interface, trained models, datasets, and SQLite database.

Model evaluation measured the performance of the sentiment classifiers on previously unseen test data. User acceptance testing was conducted with three testers, each completing nine predefined tasks. Under the agreed testing rule, the positive and negative review tasks required the correct sentiment class, not merely the display of a prediction.

5.2 Automated and Functional Test Cases

5.2.1 Automated Unit Tests

The project test suite contained five automated tests. These tests were completed successfully.

Test area	Expected result	Status
Rating-to-sentiment mapping	Ratings are correctly mapped to negative, neutral, or positive	Pass
Text cleaning	Noise is removed and important negation words are preserved	Pass
Balanced dataset sampling	The requested sample contains balanced sentiment classes	Pass
Database and feedback operations	Analyses and feedback can be created, updated, retrieved, and deleted	Pass
Invalid input validation	Invalid confidence and feedback values are rejected	Pass

5.2.2 Functional Test Cases

Test case	Expected result	Actual result	Status
Application loading	The dashboard opens successfully	Dashboard opened successfully	Pass
Category selection	The selected category is loaded	Video Games and All Beauty could be selected	Pass
Review submission	A prediction is displayed	Prediction and confidence values were displayed	Pass
Blank review	A warning is displayed	Warning was displayed	Pass
Dataset Dashboard	Charts and summary statistics are displayed	Charts and summary statistics loaded	Pass
Model Performance page	Metrics and confusion matrix are displayed	Metrics and confusion matrix loaded	Pass
Analysis history	Saved analyses are displayed	Saved analyses appeared in history	Pass
Feedback submission	Feedback is stored and displayed	Feedback was saved successfully	Pass
Clear history	Saved records are removed after confirmation	Local history was cleared after confirmation	Pass

The results indicate that the main user-interface and persistence functions operated as expected. Classification correctness was evaluated separately through model evaluation and user acceptance testing.

5.3 Model Evaluation Procedure

The evaluation used accuracy, weighted precision, weighted recall, and weighted F1-score. Accuracy measures the proportion of correctly classified reviews. Precision measures how reliable the predictions for each class are. Recall measures how many examples from each class are successfully identified. The F1-score combines precision and recall into a single measure. The All_Beauty dataset contained 38,754 training reviews and 9,689 testing reviews. The Video Games dataset contained 38,696 training reviews and 9,675 testing reviews.

5.4 Model Evaluation Results

5.4.1 Overall Model Comparison

Category	Model	Accuracy	Precision	Recall	F1-score
All Beauty	Majority baseline	33.82%	11.44%	33.82%	17.10%
All Beauty	Multinomial Naive Bayes	68.04%	68.38%	68.04%	68.18%
All Beauty	Logistic Regression	69.33%	69.26%	69.33%	69.28%
Video Games	Majority baseline	34.10%	11.63%	34.10%	17.34%
Video Games	Multinomial Naive Bayes	66.65%	67.51%	66.65%	66.87%
Video Games	Logistic Regression	68.34%	68.44%	68.34%	68.37%

Logistic Regression produced the highest accuracy and F1-score for both product categories. Compared with Multinomial Naive Bayes, Logistic Regression improved accuracy by approximately 1.29 percentage points for All Beauty and 1.70 percentage points for Video Games. Both machine-learning models performed substantially better than the Majority Class baseline. This indicates that the models learned useful patterns from the review text instead of simply predicting one sentiment category.

5.4.2 Class-Level Results

All Beauty Class Result

Sentiment class	Precision	Recall	F1-score	Support
Negative	69.54%	71.74%	70.62%	3,277
Neutral	59.74%	58.18%	58.95%	3,252
Positive	78.76%	78.29%	78.53%	3,160

Video Games Class Result

Sentiment class	Precision	Recall	F1-score	Support
Negative	70.08%	72.69%	71.36%	3,299
Neutral	58.45%	58.52%	58.48%	3,264
Positive	77.19%	74.04%	75.58%	3,112

The positive class achieved the highest F1-score in both categories. This may be because many four- and five-star reviews contain clear positive expressions. Neutral reviews were more difficult to classify because they often contain factual statements, mild opinions, or a mixture of positive and negative comments.

5.4.3 Confusion Matrices

The confusion matrices below use actual classes as rows and predicted classes as columns.

All Beauty Confusion Matrices

Actual / Predicted	Negative	Neutral	Positive
Negative	2,351	777	149
Neutral	842	1,892	518
Positive	188	498	2,474

Video Games Confusion Matrices

Actual / Predicted	Negative	Neutral	Positive
Negative	2,398	745	156
Neutral	829	1,910	525
Positive	195	613	2,304

The matrices show that most positive and negative reviews were classified correctly. The most common errors occurred between neutral and negative reviews and between neutral and positive reviews. This suggests that the model has difficulty distinguishing neutral or mixed opinions from clearly polarised opinions.

5.5 User Acceptance Testing

ID	Task	Expected result
T1	Open the application	App loads successfully
T2	Select Video Games	Correct category is selected
T3	Write a positive review	Positive prediction appears
T4	Write a negative review	Negative prediction appears
T5	Write a blank review	Validation warning appears
T6	Switch to All Beauty	Dataset and model update
T7	Open Dataset Dashboard	Charts and metrics display
T8	Open History & Feedback	Analysis history appears
T9	Submit feedback	Feedback is saved successfully

The expected result for the positive and negative review tasks was the correct sentiment class. A prediction that displayed successfully but produced the wrong class was recorded as a failed classification task. Based on the corrected testing record, the results were:

Tester	Tasks passed	Tasks failed	Overall status
Tester 1	8/9	1	Fail/Partial
Tester 2	8/9	1	Fail/Partial
Tester 3	9/9	0	Pass

The three testers completed 25 out of 27 tasks, resulting in an overall task completion rate of 92.6%. Tester 1 and Tester 2 each failed one classification task, while Tester 3 completed all nine tasks successfully. Under the strict acceptance rule, a tester must complete all nine tasks correctly to receive an overall Pass. Therefore, Tester 1 and Tester 2 are recorded as Fail/Partial, while Tester 3 is recorded as Pass

The complete task-level tester records are provided in Appendix E.

5.6 Evaluation Limitations

First, sentiment labels were derived from star ratings rather than manually annotated by human reviewers. A three-star review was labelled neutral even if the review text contained a strong positive or negative opinion. Similarly, a five-star review may still include complaints about a product feature. Second, the evaluation used one fixed 80:20 train-test split. Although the split was stratified and reproducible, a different split may produce slightly different results. Repeated cross-validation could provide a more reliable estimate in future work.

Third, the datasets contain English-language Amazon reviews from only two product categories. The results may not generalise to other product domains, languages, or social-media text. Finally, the user acceptance test involved only three classmates. The 92.6% completion rate demonstrates that the core interface tasks could be completed, but it should not be interpreted as a statistically generalisable usability result.

Overall, the system achieved useful classification performance and successfully integrated the trained models into an interactive dashboard. However, the results should be treated as decision-support information rather than as a perfect interpretation of every customer opinion.

6.0 Results and Evaluation

6.1 Achievement of Project Objectives

The first objective was to prepare approximately 50,000 Amazon reviews for the Video Games and All Beauty categories. This objective was achieved. After duplicate and empty-record removal, the final datasets contained 48,371 Video Games reviews and 48,443 All Beauty reviews. The second objective was to preprocess the review data and assign sentiment labels. This was achieved through rating-based labelling, HTML and URL removal, lowercasing, stop-word removal, negation preservation, and lemmatisation. Reviews with missing or empty text were excluded.

The third objective was to develop category-specific sentiment-classification models. This was achieved using TF-IDF features and Logistic Regression. Separate vectorizers and classifiers were trained for the Video Games and All Beauty categories. The fourth objective was to compare and evaluate the models. This was achieved using a Majority Class baseline, Multinomial Naive Bayes, and Logistic Regression. Logistic Regression achieved the strongest performance for both categories, with an accuracy of 69.33% for All Beauty and 68.34% for Video Games.

The fifth objective was to develop an interactive sentiment-analysis dashboard. This was achieved through the Streamlit application. Users can select a category, analyse a review, view confidence scores, explore sentiment distributions, and inspect model-performance results. The sixth objective was to provide analysis history and feedback functionality. This was achieved using a local SQLite database. The application stores submitted reviews, predicted sentiment, confidence, timestamps, and optional user feedback.

The seventh objective was to evaluate usability through user testing. This was achieved using three testers who each completed nine tasks. A total of 25 out of 27 tasks were completed successfully, giving a completion rate of 92.6%. The results show that the main dashboard functions were usable, although difficult language caused two classification-related failures.

6.2 Discussion of Model Results

Logistic Regression was the best-performing tested model for both product categories. For All Beauty, it achieved 69.33% accuracy and a weighted F1-score of 69.28%. For Video Games, it achieved 68.34% accuracy and a weighted F1-score of 68.37%.

The Logistic Regression model performed better than the Multinomial Naive Bayes model by approximately 1.29 percentage points in All Beauty accuracy and 1.70 percentage points in Video Games

accuracy. Both models performed considerably better than the Majority Class baseline, which achieved approximately one-third accuracy because the datasets contained three approximately balanced sentiment classes.

The All Beauty model performed slightly better than the Video Games model. This difference may be related to the vocabulary and writing style used in each product category. Reviews in different domains may use different expressions, technical terms, and patterns of complaint or praise.

The class-level results showed that positive reviews were classified most successfully. The positive class achieved an F1-score of 78.53% for All Beauty and 75.58% for Video Games. This may be because positive reviews often contain clear expressions such as “excellent”, “love”, “great”, or “highly recommended”.

Neutral reviews were the most difficult class to identify. The neutral F1-score was 58.95% for All Beauty and 58.48% for Video Games. Neutral reviews may contain factual descriptions, mild opinions, or both positive and negative statements. For example, a reviewer may praise a product’s design but criticise its durability. TF-IDF and Logistic Regression may have difficulty determining which part of the review represents the overall opinion.

The confusion matrices also showed that neutral reviews were frequently confused with negative and positive reviews. This demonstrates that the model is more effective at identifying strongly positive or strongly negative language than ambiguous or mixed opinions.

6.3 Challenges Encountered

One major challenge was handling the size of the raw Amazon review data. Large JSONL files cannot be processed as easily as small spreadsheet files. Reservoir sampling was used to select a reproducible, approximately balanced sample without storing the entire source file in memory.

Another challenge involved text quality. Reviews contained HTML tags, URLs, punctuation, encoding artefacts, spelling variations, and inconsistent formatting. These elements had to be removed or normalised before the text could be used for training.

Stop-word removal also required careful consideration. Removing every stop word could remove negation terms that change sentiment meaning. Words such as “not”, “no”, “never”, and “nothing” were therefore retained.

The rating-based labelling process created another challenge. A star rating represents an overall evaluation, but it does not always match the exact sentiment expressed in the review text. A five-star review may contain a complaint, while a one-star review may praise one feature of a product. This explains some of the model’s classification errors.

A further challenge was interpreting performance fairly. Accuracy alone did not show which sentiment classes were difficult to identify. Precision, recall, F1-score, classification reports, and confusion matrices were therefore included to provide a more detailed evaluation.

Finally, user testing showed that successful interface operation does not guarantee perfect language interpretation. Long mixed reviews, sarcasm, slang, abbreviations, and informal expressions can result in unexpected predictions.

6.4 Limitations of the System

The main limitation is that sentiment labels were derived from star ratings rather than independent human annotations. The evaluation therefore measures how well the model reproduces the rating-based labelling rule. It does not perfectly measure whether the model understands the complete meaning of each review. The use of TF-IDF also limits the system's contextual understanding. TF-IDF represents important words and short phrases but does not fully understand word order, sarcasm, humour, slang, or relationships between different parts of a long review.

The system is trained only on English reviews from the Video Games and All Beauty categories. Its performance may be different when applied to other product categories, languages, social-media posts, or customer-service messages.

The evaluation was based on one fixed stratified 80:20 train-test split. Although this procedure was reproducible, repeated cross-validation could provide a more reliable estimate of model performance. The current project also did not perform extensive hyperparameter tuning.

The confidence values displayed by the Logistic Regression model should not be interpreted as guaranteed probabilities of correctness. A high-confidence prediction may still be incorrect, especially when a review contains sarcasm, mixed sentiment, or vocabulary that is not well represented in the training data.

The SQLite database is local to the application instance and is not designed as a production multi-user database. On a cloud deployment, records may be shared between users or lost after a restart or redeployment. The current application also does not provide user authentication, encryption, or permanent cloud storage.

The usability test involved only three classmates. The 25/27 result demonstrates that most core tasks could be completed, but it cannot be generalised to all users. A larger study would be required to measure usability more reliably.

6.5 Ethical and Practical Considerations

Users should avoid entering confidential, personal, or sensitive information into the review box because submitted reviews may be stored in the local analysis history database.

The dashboard is intended as an educational and decision-support application. It should not be used to automatically remove reviews, penalise customers, or make high-stakes decisions about individuals.

The use of star ratings as sentiment labels should be clearly explained to users. A model prediction and confidence value provide an automated estimate, but the final interpretation should remain with a human reviewer.

Future deployment with real customer data would require stronger privacy controls, user authentication, access permissions, data-retention policies, and a secure external database.

Overall, the project achieved its main objectives by producing a working three-class sentiment-analysis dashboard. The results demonstrate that a transparent TF-IDF and Logistic Regression approach can provide useful predictions with modest computational requirements. However, the limitations also show that automated sentiment analysis should be used to support human review rather than replace it.

7.0 Conclusion

This project developed an interactive AI Sentiment Analysis Dashboard for Amazon product reviews. The system focuses on the Video Games and All Beauty categories and classifies reviews into three sentiment classes: positive, neutral, and negative.

The project completed the main stages of an AI application lifecycle. Review data was collected and prepared, star ratings were converted into sentiment labels, and text preprocessing was performed to remove unnecessary content. TF-IDF was then used to transform the review text into numerical features. Category-specific Logistic Regression models were trained and compared with a Majority Class baseline and Multinomial Naive Bayes.

The Logistic Regression model achieved the strongest performance for both categories. It achieved an accuracy of 69.33% and a weighted F1-score of 69.28% for All Beauty. For Video Games, it achieved an accuracy of 68.34% and a weighted F1-score of 68.37%. These results were higher than the baseline and Naive Bayes models, showing that the selected approach learned useful patterns from the review text.

The trained models were integrated into a Streamlit dashboard. Users can select a product category, enter a review, view a predicted sentiment, inspect confidence values, explore dataset distributions, and view model-performance results. The application also includes SQLite-based analysis history and feedback functionality.

Automated testing confirmed that the main data-preparation and database functions operated correctly, with five tests passing. User acceptance testing involved three testers completing nine tasks each. A total of 25 out of 27 tasks were completed successfully, resulting in a task completion rate of 92.6%. The two failed tasks involved difficult language, including mixed opinions, slang, or abbreviations.

The project achieved its main objectives within the defined scope. However, the results also demonstrate that sentiment analysis is not always straightforward. The use of star ratings as proxy labels, the limitations of TF-IDF, and the difficulty of understanding sarcasm and mixed reviews mean that the dashboard should be used as a decision-support tool rather than as a replacement for human interpretation.

7.1 Future Recommendations

First, the sentiment labels could be improved using manually annotated reviews. Human annotation would provide a stronger measure of the sentiment expressed in the text and reduce the limitations of relying only on star ratings.

Second, more advanced models could be evaluated. Transformer-based models such as BERT or RoBERTa may provide better contextual understanding of long reviews, sarcasm, and mixed opinions. Their performance should be compared with the current Logistic Regression model using the same datasets and evaluation procedure.

Third, aspect-based sentiment analysis could be added. Instead of producing only one overall sentiment label, the system could identify opinions about specific aspects such as product quality, price, delivery, design, comfort, or usability.

Fourth, the text-processing pipeline could be improved to handle slang, abbreviations, spelling errors, emojis, sarcasm, and multilingual reviews. These features are common in real customer feedback and may improve the system's ability to process informal language.

Fifth, model evaluation could be strengthened through repeated cross-validation, systematic hyperparameter tuning, confidence calibration, and additional error analysis. These techniques would provide a more reliable estimate of performance and help identify the types of reviews that cause incorrect predictions.

Sixth, the application database could be migrated from local SQLite storage to a secure external database. Authentication, access control, encryption, and data-retention policies should be added before the system is used with real customer data.

Seventh, the dashboard could be extended with batch review analysis, tested CSV uploading, downloadable reports, trend analysis over time, and filtering by rating or verified-purchase status. These features should only be added after implementation and testing.

Finally, future usability evaluation should involve more participants from different backgrounds and levels of technical experience. Accessibility improvements, mobile testing, and a standardised usability questionnaire could also provide more reliable evidence about user satisfaction.

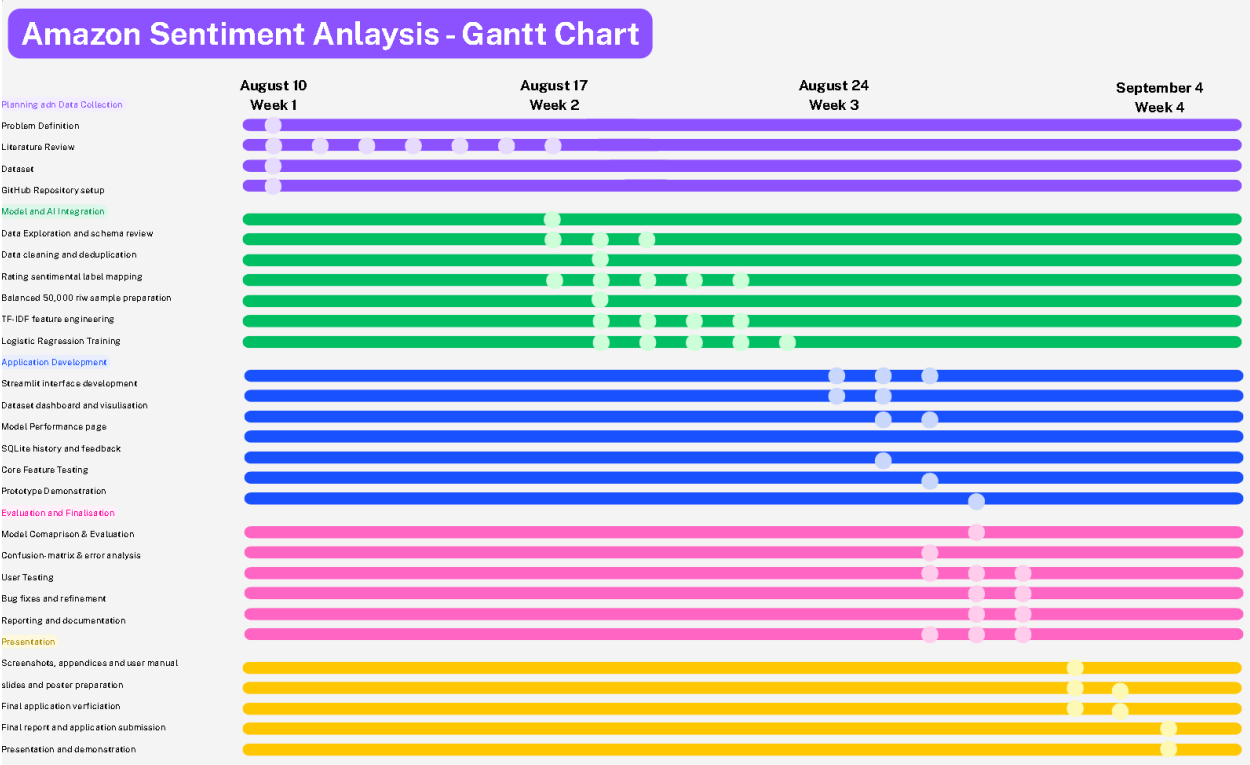
The current project provides a reproducible foundation for future sentiment-analysis development. Its combination of category-specific datasets, transparent machine-learning models, visual analysis, and user feedback creates a practical starting point for more advanced customer-opinion systems.

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Appendices

Appendix A: Project Schedule (Gantt Chart)



Appendix B: Source Code

<https://github.com/minerb388-wq/AI-Sentiment-Analysis>

Appendix C: Dataset Samples

<https://amazon-reviews-2023.github.io/>

Appendix D: Application

<https://ai-sentiment-analysis-zz8emsybtgbi8rj3ltc57s.streamlit.app/#review-analyzer>

Appendix E: Tester Result Table**Tester 1:**

Task	Expected result	Result	Status	Remark
T1: Open app	Application loads	Application loaded successfully	Pass	No issue
T2: Select Video Games	Video Games is selected	Video Games selected	Pass	No issue
T3: Positive review	Prediction is displayed	Neutral prediction displayed	Failed/ Partially	Long mixed reviews may be classified as neutral
T4: Negative review	Prediction is displayed	Negative prediction displayed	Pass	When adding sarcasm, it understood it as positive
T5: Blank review	Warning is displayed	Warning displayed	Pass	No issue
T6: Switch to All Beauty	Category and model change	All Beauty loaded	Pass	No issue
T7: Open dashboard	Charts are displayed	Charts displayed correctly	Pass	No issue
T8: Open history	Saved analysis appears	Analysis appeared in history	Pass	No issue
T9: Submit feedback	Feedback is saved	Feedback saved successfully	Pass	No issue

Tester 2:

Task	Expected result	Result	Status	Remark
T1: Open app	Application loads	Application loaded successfully	Pass	Opened on phone
T2: Select Video Games	Video Games is selected	Video Games selected	Pass	No issues
T3: Positive review	Prediction is displayed	Positive prediction displayed	Pass	Can understand offensive words to exaggerate positive
T4: Negative review	Prediction is displayed	Negative prediction displayed	Failed	Does not understand slangs or abbreviations

T5: Blank review	Warning is displayed	Warning displayed	Pass	No issue
T6: Switch to All Beauty	Category and model change	All Beauty loaded	Pass	No issue
T7: Open dashboard	Charts are displayed	Charts displayed correctly	Pass	No issue
T8: Open history	Saved analysis appears	Analysis appeared in history	Pass	No issue
T9: Submit feedback	Feedback is saved	Feedback saved successfully	Pass	No issue

Tester 3:

Task	Expected result	Result	Status	Remark
T1: Open app	Application loads	Application loaded successfully	Pass	No issue
T2: Select Video Games	Video Games is selected	Video Games selected	Pass	No issue
T3: Positive review	Prediction is displayed	Positive prediction displayed	Pass	Clear result
T4: Negative review	Prediction is displayed	Negative prediction displayed	Pass	No issue
T5: Blank review	Warning is displayed	Warning displayed	Pass	No issue
T6: Switch to All Beauty	Category and model change	All Beauty loaded	Pass	No issue
T7: Open dashboard	Charts are displayed	Charts displayed correctly	Pass	No issue
T8: Open history	Saved analysis appears	Analysis appeared in history	Pass	No issue
T9: Submit feedback	Feedback is saved	Feedback saved successfully	Pass	No issue